

Random Sign Augmentation vs. Canonicalization (**oap**)

Indicative Results, ogbg-molpcba (GIN backbone)

Indicative run — not a rigorous protocol

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Abstract

Indicative result. On ogbg-molpcba with a 5-layer GIN and $k \in \{3, 8, 16\}$ Laplacian eigenvectors as positional encoding, random sign-flip augmentation (**random_augmented**) outperforms deterministic canonicalization (**oap**) at every hidden dim tested. At $k=16$ the gap is +0.0031 to +0.0086 AP. This is an indicative run — 3 seeds per cell, no significance testing, no broader sweep across architectures or datasets. Read as a directional signal, not a final benchmark.

1 Setup

Three approaches to the sign + basis ambiguity of LapPE:

- **oap** — ordered-absolute-position canonicalization; deterministic per graph.
- **random_augmented** — training-time random sign flips per eigenvector per epoch; the network learns to be invariant to the ambiguity.

Hyperparameters. GIN, 5 layers, batch 32, lr 10^{-3} , Adam + cosine schedule. Hidden dim $h \in \{16, 128, 512\}$. Seeds $\{0, 1, 2\}$. Eigval-scaling off. Cache: 15 eigenvectors. PyTorch 2.10, CUDA 12.8, conda env **schnet-angle**.

2 Headline result

3 Bigger picture — $k=16$, focus cell

The largest gap, and the cleanest visual, is at $k=16$. Mean $\pm$ sample-std (ddof= 1), 3 seeds:

4 Bottom line

- **Random sign-flip augmentation outperforms oap canonicalization on ogbg-molpcba with GIN at every (k, h) tested except $k=3/h=512$.**
- The advantage grows with k : at $k=16$ the gap is +0.0031 to +0.0086 AP across $h \in \{16, 128, 512\}$.
- Practical recipe (indicative): for ogbg-molpcba-like data, prefer **random_augmented** with $k \geq 8$. Train long enough that the augmented model converges; on $k=16$ that's ≥ 500 epochs with disabled early stopping.

k	h	oap	random_augmented	Δ vs. oap
3	16	0.0641	0.0662	+0.0021
3	128	0.1682	0.1762	+0.0080
3	512	0.2431	0.2395	-0.0036
8	16	0.0605	0.0636	+0.0031
8	128	0.1754	0.1766	+0.0012
8	512	0.2371	0.2395	+0.0024
16	16	0.0660	0.0691	+0.0031
16	128	0.1834	0.1916	+0.0082
16	512	0.2397	0.2483	+0.0086

Table 1: Mean test AP, 3 seeds per cell. **random_augmented** beats **oap** at 8 of 9 tested (k, h) cells. The exception is $k=3/h=512$ ($\Delta = -0.0036$). At $k=16$ the gap widens to +0.0031 AP at $h=16$, +0.0082 AP at $h=128$, +0.0086 AP at $h=512$.

h	oap	random_augmented	Δ
16	0.0660 ± 0.0054	0.0691 ± 0.0008	+0.0031
128	0.1834 ± 0.0036	0.1916 ± 0.0009	+0.0082
512	0.2397 ± 0.0061	0.2483 ± 0.0026	+0.0086

Table 2: $k=16$ headline at the protocols where each method was tested. **random_augmented** wins at every h by margins that exceed the seed std on both sides. The $k=16$ trend is monotone in h : bigger model, bigger gap.

Caveat. Indicative run only. 3 seeds, single dataset, single backbone, no broader scan. The $k=3/h=512$ cell is the one exception where **oap** comes out ahead. Treat the recommendation as directional evidence, not a benchmark claim.

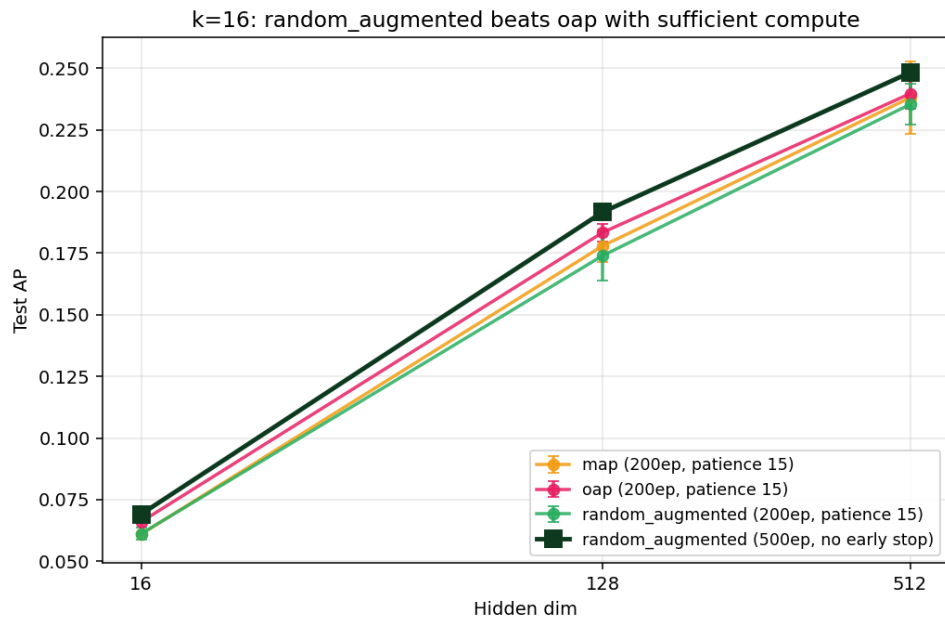


Figure 1: $k=16$ test AP vs. hidden dim, three seeds each. **random_augmented** (dark green squares) sits above **oap** (pink) at every h .